MUSICANOTE · AI Arrangement

An Explainable Arrangement Engine

It analyses a single-line MusicXML melody, decides the meaning and function of every passage, and builds a four-stage arrangement on those decisions. Every note traces back to the analysis and its source.

Pipeline

1 Analysis

Local keys · metre and pickups · phrases (2–4–8-bar hierarchy) · motifs and variants · harmony inference · form and tension

2 Semantic Map

Form role, harmonic function, cadences and climax for every bar and passage

3 Plan

Intensity, texture, register and instrument roles per phrase

4 Realisation

Accompaniment retrieval (best path across bars) · role-based pitch transfer · voice leading (7ths, leading tones resolve) · difficulty limits

5 Validation

Melody preserved · hand span and chord size · voice completeness · corpus statistics

Data Foundations

654
Movements
DCML expert harmony annotations (~130,000): transitions, cadences, harmonic rhythm

392
Songs
Five-edition set (melody + four ensembles): harmony ground truth, role statistics

77,700
Bars
Accompaniment library from 924 arrangers' piano scores

80 / 20
Split
Train / evaluation split by song; evaluation songs never feed extraction

Roles by Stage (Corpus Targets)

1 Piano Solo
Melody kept; octave doubling (44%) and chord fill by intensity; texture over a planned bass

2 + Violin
Takes the melody (19 of 30 songs); varied on repeats, e.g. an octave up

3 + Cello
Mainly the bass line (pitch class matches the piano bass ~75%); links where the melody rests

4 + Viola
Inner voice and countermelody; sustained 3rd/7th guide tones and answers

Controls

- Difficulty 1–10: hand span, chord size and left-hand rhythm limits (Chinese piano grades)
- Harmony: given chords / inferred from the melody / newly generated (pop, jazz, classical; jazz per beat)
- Style plan by passage, re-arrange selected bars only, alternate arrangements
- Web workbench: score rendering and playback, side by side with the original

Explainability

- Every note records the stage and source that made it (library song, bar, role)
- Generated harmony is labelled a generation assumption, never passed off as the composer's
- Output: five MusicXML files per song (analysis lead sheet + four ensembles)

Explore the Site

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